

29/04/2024



Aakash

Medical | IIT-JEE | Foundations

Corporate Office: Aakash Tower, 8, Pusa Road, New Delhi-110005, Ph.011-47623456

Phase-I
CODE-A

FINAL TEST SERIES for NEET-2024

MM : 720

Test -12

Time : 3 Hrs. 20 Mins.

Mock Test for NEET (Complete Syllabus of class XI & XII)

Instructions :

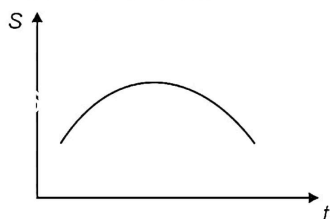
- There are two sections in each subject, i.e. Section-A & Section-B. You have to attempt all 35 questions from Section-A & only 10 questions from Section-B out of 15.
- Each question carries 4 marks. For every wrong response 1 mark shall be deducted from the total score. Unanswered / unattempted questions will be given no marks.
- Use blue/black ballpoint pen only to darken the appropriate circle.
- Mark should be dark and completely fill the circle.
- Dark only one circle for each entry.
- Dark the circle in the space provided only.
- Rough work must not be done on the Answer sheet and do not use white-fluid or any other rubbing material on the Answer sheet.

PHYSICS

Choose the correct answer :

SECTION - A

1. Position-time($S-t$) graph of a particle moving in a straight line is shown below.



Speed of the particle is

- Constant
 - First increases then decreases
 - First decreases then increases
 - Continuously increases
2. A stone is released from the top of a tower and it reaches ground in 8 s. Its height from the ground when its speed is 40 m s^{-1} is ($g = 10 \text{ m/s}^2$)
- 80 m
 - 240 m
 - 160 m
 - 120 m

3. An object of mass 3 kg is moving with 36 km/h is acted by a force and its velocity becomes 54 km/h in the same direction of motion. Impulse of the force is

- Zero
- 75 kg m s^{-1}
- 25 kg m s^{-1}
- 15 kg m s^{-1}

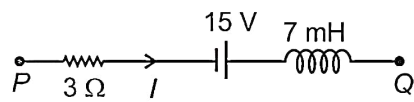
4. A piston of cross-sectional area 150 cm^2 is used in a hydraulic lift to exert a force of 10^6 dynes on the water. The cross-sectional area of the other piston is $15 \times 10^5 \text{ cm}^2$. The load it balance is

- 10^4 N
- 10^5 N
- 10^3 N
- 10^6 N

5. 2 moles of diatomic gas is contained in a container of volume V has pressure P . Its internal energy is

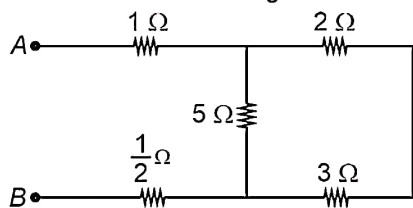
- $\frac{3}{2}PV$
- $3PV$
- $\frac{5}{2}PV$
- $5PV$

Test-12 (Code-A)

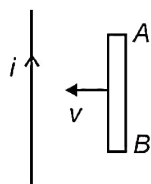
6. Two charges $+3Q$ and $-Q$ are placed at some distance apart. If N is the number of electric field lines originates from $+3Q$, then the number of field lines which terminates at infinity are
- (1) $\frac{N}{3}$ (2) $3N$
 (3) $\frac{2N}{3}$ (4) $\frac{N}{3}$
7. An electron and a proton are moving such that they have same de-Broglie wavelength. Then the kinetic energy of the proton is
- (1) Equal to the kinetic energy of electron
 (2) Greater than the kinetic energy of electron
 (3) Smaller than the kinetic energy of electron
 (4) Infinite
8. If energy of a body moving along x-axis at time 't' and position 'x' is given by $E = \frac{A - x^2}{Bt}$, then the dimensions of $\left[\frac{A}{B}\right]$ will be
- (1) $[ML^2 T^{-1}]$ (2) $[ML T^{-2}]$
 (3) $[M^{-1} L^3 T^{-2}]$ (4) $[M^{-1} L^2 T]$
9. A particle is projected from ground at an angle of 60° from the horizontal with speed 150 m/s. The time (in second) after which the velocity of particle makes an angle 90° with its initial velocity is
- (1) $10\sqrt{3}$
 (2) $\sqrt{30}$
 (3) $\sqrt{90}$
 (4) $\sqrt{150}$
10. Young's modulus of a metal wire depends upon
- (1) Length of the wire
 (2) Area of cross-section of the wire
 (3) Volume of the wire
 (4) Nature of it's material
11. Four charges, each equal to $-Q$ are placed at the four corners of a regular pentagon and charge q is at its centre. Now a charge $+Q$ is placed on the fifth corner. If r is the distance of centre to each corner then the magnitude of net force on charge q is
- (1) $\frac{2KQq}{r^2}$ (2) $\frac{KQq}{r^2}$
 (3) $\frac{KQq}{2r^2}$ (4) Zero
12. The network shown below is a part of a complete circuit. If a current of 5 A is flowing through it and it is increasing at a rate of 2×10^3 A/s, then magnitude of potential difference between the points P & Q will be
- 
- (1) 20 V (2) 15 V
 (3) 7 V (4) 14 V
13. A hollow conducting sphere of radius R has a charge Q . The electric potential (V) due to the sphere at a distance r from the centre of sphere is
- (1) $V = 0$ for $r < R$ and $V = \frac{Q}{4\pi\epsilon_0 r}$ for $r \geq R$
 (2) $V = \frac{Q}{4\pi\epsilon_0 r}$ for all values of r
 (3) $V = \frac{Q}{4\pi\epsilon_0 R}$ for $r < R$
 and $V = \frac{Q}{4\pi\epsilon_0 r}$ for $r \geq R$
 (4) $V = \frac{2Q}{\pi\epsilon_0 r}$ for $r < R$ and $V = \frac{Q}{4\pi\epsilon_0 r}$ for $r \geq R$

Space for Rough Work

14. The equivalent resistance between A and B for the network shown in the figure is



- (1) $2\ \Omega$ (2) $3\ \Omega$
 (3) $4\ \Omega$ (4) $7\ \Omega$
15. To decrease the current sensitivity of a moving coil galvanometer, one can decrease
- (1) Number of turns in coil
 (2) Magnetic field inside it
 (3) Area of its coil
 (4) All of the above
16. Rotational kinetic energy per molecule for a diatomic gas sample kept at temperature T is (k_B is Boltzmann constant)
- (1) $\frac{1}{2}k_B T$ (2) $\frac{3}{2}k_B T$
 (3) $k_B T$ (4) $\frac{5}{2}k_B T$
17. When a sound wave is refracted from air to water, then its
- (1) Wavelength increases
 (2) Frequency decreases
 (3) Wave number increases
 (4) Wave velocity remain unchanged
18. A current carrying long wire and a conducting rod AB are in the same plane. If the rod moves toward the wire with constant speed v as shown, then



- (1) End A will be at lower potential with respect to B
 (2) A and B are at same potential
 (3) End A will be at higher potential with respect to B
 (4) Both end A and B are at zero potential

19. The electric potential $V(x)$ in a region around the origin is given by $V(x) = 4x$ volts. Net electric charge enclosed in a cube of 1 m side with its centre at the origin, will be

- (1) $8\epsilon_0$ (2) $-4\epsilon_0$
 (3) $-8\epsilon_0$ (4) Zero

20. In an AC circuit the instantaneous values of emf and current are :

$$E = 200 \sin(314t) \text{ volt}$$

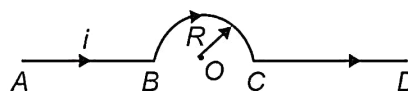
&

$$I = 10 \sin\left(314t + \frac{\pi}{3}\right) \text{ amp}$$

The average power consumed in watt is

- (1) 2000 (2) 1000
 (3) 1500 (4) 500

21. In the figure shown the magnetic field at the centre (O) of the arc due to the current in section CD will be



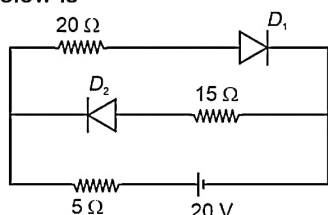
- (1) $\frac{\mu_0 i}{2R}$ (2) $\frac{\mu_0 i}{4R}$
 (3) Zero (4) $\frac{\mu_0 i}{8R}$

22. Correct volume-temperature relationship for an ideal gas undergoing adiabatic change is ($\gamma = C_p/C_v$)

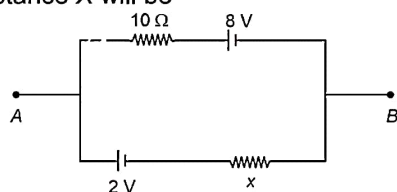
- (1) $V^{\gamma-1} T^\gamma = \text{constant}$ (2) $VT^\gamma = \text{constant}$
 (3) $V^{\gamma-1} T = \text{constant}$ (4) $V^{1-\gamma} T = \text{constant}$

Space for Rough Work

23. The current I through $20\ \Omega$ resistor in the circuit given below is



- (1) 0.4 A (2) 0.2 A
(3) 1 A (4) 0.8 A
24. If $V_A - V_B = 6\text{ V}$ in the given figure then resistance X will be



- (1) $10\ \Omega$ (2) $5\ \Omega$
(3) $6\ \Omega$ (4) $20\ \Omega$
25. If the resistance of a galvanometer coil is R , then the shunt resistance required to convert it into an ammeter of range 3 times, will be

- (1) $\frac{R}{4}$ (2) $\frac{R}{2}$
(3) $4R$ (4) $2R$

26. A body takes 10 minutes to cool from 90°C to 80°C . if the temperature of the surrounding is 20°C . Time taken by it to cool from 80°C to 70°C in same surrounding will be

- (1) 10 minutes (2) 5 minutes
(3) 11.8 minutes (4) 15.4 minutes

27. The amount of heat required to convert 10 g of ice at 0°C to water at 80°C will be

- (1) 800 cal (2) 2400 cal
(3) 1600 cal (4) 3200 cal

28. If wavelength of given light waves in air and in a medium are $5000\ \text{\AA}$ and $2500\ \text{\AA}$ respectively, then the critical angle at the interface of medium air is

- (1) 30° (2) 45°
(3) 60° (4) 37°

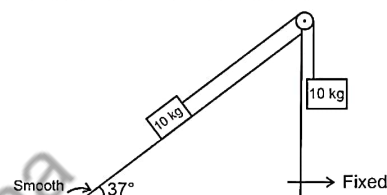
29. A simple microscope consists of a convex lens of power $+10\text{ D}$. If the image is formed at infinity then the magnifying power is

- (1) 3.5 (2) 5
(3) 4 (4) 2.5

30. A ray of light strikes a material's slab at an angle of incidence 53° . If the reflected and refracted rays are perpendicular to each other, then refractive index of the material is

- (1) $\frac{3}{5}$ (2) $\frac{4}{3}$
(3) $\frac{4}{5}$ (4) $\frac{5}{4}$

31. In the figure shown below the wedge is smooth and fix. If the given system is released from rest, then acceleration of the blocks is



- (1) 1 m/s^2 (2) 4 m/s^2
(3) 3 m/s^2 (4) 2 m/s^2

32. A particle performs SHM on x -axis with amplitude A and time period T . If at $t = 0$ particle is at mean position and moving towards positive x -axis, then time taken by it to travel a distance $\frac{A}{2}$ will be

- (1) $\frac{T}{6}$ (2) $\frac{T}{4}$
(3) $\frac{T}{8}$ (4) $\frac{T}{12}$

33. The radius of gyration of a disc of mass 500 g and radius 5 cm about an axis passing through its centre and perpendicular to its plane, is

- (1) 5 cm (2) 2.5 cm
(3) 3.54 cm (4) 1.51 cm

Space for Rough Work

34. A system consists of 3 particles each of mass m located at point (1, 2), (2, 3) and (3, 4). The coordinates of the center of mass of system are

(1) (1, 2) (2) (3, 2)
(3) (2, 3) (4) (2, 1)

35. Given below are two statements : one is labelled Assertion (A) and other is labelled as Reason (R).

Assertion (A) : Time varying magnetic field induces electric field.

Reason (R) : Time varying electric field never induces magnetic field.

In the light of the above statements, choose the correct answer.

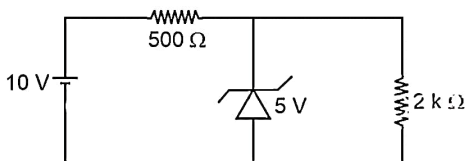
- (1) Both **A** and **R** are true and **R** is the correct explanation of **A**
(2) Both **A** and **R** are true and **R** is **NOT** the correct explanation of **A**
(3) **A** is true but **R** is false
(4) Both **A** and **R** are false

SECTION - B

36. A long straight thin wire is placed along negative z direction starting from origin and it carries a current I . The magnitude of magnetic field at point 'P' having coordinates (0, 0, 5 m) is

(1) Zero (2) $\frac{\mu_0 I}{10\pi}$ T
(3) $\frac{\mu_0 I}{5\pi}$ T (4) $\frac{\mu_0 I}{20\pi}$ T

37. In the following circuit, the current flowing through the zener diode will be



(1) 2.5 mA (2) 5 mA
(3) 2.0 mA (4) 7.5 mA

38. One mole of an ideal monoatomic gas is expanded according to the law $\frac{P}{V} = \text{constant}$. If the temperature of the gas increased from T_0 to $2T_0$, then heat supplied to the gas in this process is

(1) $2RT_0$ (2) $\frac{3}{2}RT_0$
(3) RT_0 (4) $\frac{RT_0}{2}$

39. Two independent light sources of same frequency and intensities I_1 and I_2 superimpose. The maximum intensity after superposition will be

(1) $I_1^2 + I_2^2$ (2) $I_1 + I_2$
(3) $(\sqrt{I_1} + \sqrt{I_2})^2$ (4) $(\sqrt{I_1} - \sqrt{I_2})^2$

40. A 10 V ac supply is applied to a circuit consisting of a resistance and an inductive reactance of $5\sqrt{3} \Omega$ in series. If the voltage across the resistor is 5 V, then the current in the inductor is

(1) 1 A (2) 2 A
(3) 5 A (4) $\frac{1}{\sqrt{3}}$ A

41. For an equilateral prism, the condition of minimum deviation is achieved when angle of incidence is equal to 45° . Refractive index of the material of prism is

(1) $\sqrt{3}$ (2) $\sqrt{2}$
(3) $2\sqrt{3}$ (4) $2\sqrt{2}$

42. Part AB of the track shown below is smooth and beyond point B the horizontal track is rough. A block of mass 2 kg is released from rest at point A and it stops after travelling 8 m on track BC. The value of friction coefficient μ between block and track BC is

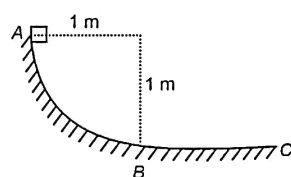
Space for Rough Work

(1) $\frac{1}{4}$

(2) $\frac{1}{8}$

(3) $\frac{3}{4}$

(4) $\frac{6}{5}$



43. The focal length of a plano-convex lens is 20 cm and the radius of curvature of curved surface is $\frac{40}{3}$ cm, then the refractive index of the material of lens is

(1) $\frac{3}{2}$

(2) $\frac{5}{3}$

(3) $\frac{4}{3}$

(4) $\frac{5}{2}$

44. Given below are two statements: one is labelled Assertion (A) and other is labelled as Reason (R).

Assertion (A): By de-Broglie hypothesis, $\lambda = \frac{h}{p}$

is valid for both the electron and the photon.

Reason (R): If the electron has the same wavelength as a photon then they have same energy.

In the light of the above statements, choose the correct answer.

- (1) Both **A** and **R** are true and **R** is the correct explanation of **A**
- (2) Both **A** and **R** are true and **R** is **NOT** the correct explanation of **A**
- (3) **A** is true but **R** is false
- (4) Both **A** and **R** are false

45. The maximum speed of a particle executing SHM is 2 m s^{-1} and maximum acceleration is $\pi \text{ m s}^{-2}$. Its frequency is

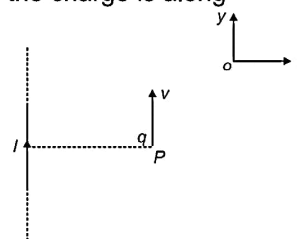
(1) 0.25 s^{-1}

(2) 2 s^{-1}

(3) 4 s^{-1}

(4) 0.5 s^{-1}

46. A very long straight wire carries a current I is along y -axis. At the instant when a charge $+q$ at point P has velocity v as shown, the direction of force on the charge is along



- (1) Negative x -axis
- (2) Positive x -axis
- (3) Negative z -axis
- (4) Positive z -axis

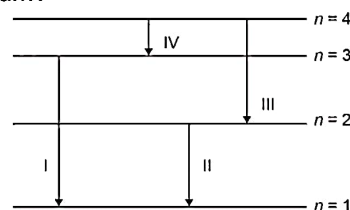
47. In a fusion process, two nuclei A and B fuse to form nucleus C , their binding energies being E_a , E_b and E_c respectively. Then

- (1) $E_c < E_a + E_b$
- (2) $E_c = E_a + E_b$
- (3) $E_c > E_a + E_b$
- (4) $E_c = 2E_a$

48. Name the gate which represents the Boolean expression $y = \bar{A} + \bar{B}$

- (1) NOR
- (2) NAND
- (3) OR
- (4) AND

49. The diagram shows the energy levels for an electron transition in a certain amount of hydrogen gas atom. For which transition wavelength of the emitted photon will be minimum?



- (1) I
- (2) II
- (3) III
- (4) IV

50. A body has weight 1000 N on the surface of the earth. How much would it weigh at 1600 km above the surface of the earth?

(Given radius of earth is 6400 km and $g = 10 \text{ ms}^{-2}$)

- (1) 760 N
- (2) 640 N
- (3) 320 N
- (4) 580 N

Space for Rough Work

SECTION - A

51. Match List-I with List-II

	List-I (Quantum number)		List-II (Orbital)
(a)	$n = 3, l = 2$	(i)	$2p$
(b)	$n = 3, l = 1$	(ii)	$3d$
(c)	$n = 2, l = 1$	(iii)	$3p$
(d)	$n = 4, l = 3$	(iv)	$4f$

Choose the correct answer from the options given below.

- (1) (a)-(ii), (b)-(iii), (c)-(i), (d)-(iv)
 (2) (a)-(iii), (b)-(ii), (c)-(iv), (d)-(i)
 (3) (a)-(ii), (b)-(i), (c)-(iii), (d)-(iv)
 (4) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)

52. Match List-I with List-II

	List-I Oxide		List-II Nature
(a)	Cl_2O_7	(i)	Basic
(b)	Al_2O_3	(ii)	Acidic
(c)	Na_2O	(iii)	Neutral
(d)	N_2O	(iv)	Amphoteric

Choose the correct answer from the options given below.

- (1) (a)-(iii), (b)-(i), (c)-(ii), (d)-(iv)
 (2) (a)-(ii), (b)-(i), (c)-(iii), (d)-(iv)
 (3) (a)-(ii), (b)-(iv), (c)-(i), (d)-(iii)
 (4) (a)-(iv), (b)-(ii), (c)-(i), (d)-(iii)

53. A first order reaction is 30% complete in 3.32 seconds.

The half life of a reaction is $\left(\log \frac{10}{7} = 0.155 \right)$

- (1) 7.65 seconds (2) 5.45 seconds
 (3) 8.95 seconds (4) 6.45 seconds

54. Incorrect statement about the characteristics of interstitial compounds is

- (1) They have higher melting points than parent metals
 (2) They are very hard, some borides approach diamond in hardness
 (3) They retain metallic conductivity
 (4) They are chemically more reactive than parent metals

55. If $\Delta_f H^\circ$ of $\text{CaCO}_3(\text{s})$, $\text{CaO}(\text{s})$ and $\text{CO}_2(\text{g})$ are -1300 , -550 and -500 kJ/mol respectively then $\Delta_r H^\circ$ of the reaction will be

- (1) -150 kJ/mol (2) 150 kJ/mol
 (3) -2450 kJ/mol (4) 2450 kJ/mol

56. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A): Λ_m for weak electrolytes shows a sharp decrease when the electrolytic solution is diluted.

Reason (R): For weak electrolytes degree of dissociation increases with dilution of solution.

In the light of above statements choose the most appropriate answer from options given below.

- (1) Both (A) and (R) are true and (R) is correct explanation of (A)
 (2) Both (A) and (R) are true but (R) is not correct explanation of (A)
 (3) Only (A) is true and (R) is false
 (4) Only (A) is false and (R) is true

57. In which of the following options, the order of arrangement does not agree with the variation of property indicated against it?

- (1) $\text{I} < \text{Br} < \text{F} < \text{Cl}$ (negative electron gain enthalpy)
 (2) $\text{Cl}^- < \text{S}^{2-} < \text{Ca}^{2+} < \text{K}^+$ (increasing ionic size)
 (3) $\text{B} < \text{C} < \text{O} < \text{N}$ (first ionisation enthalpy)
 (4) $\text{C} < \text{N} < \text{O} < \text{F}$ (electronegativity)

Space for Rough Work

58. If radius of third Bohr orbit of the He^+ ion is 238.05 pm. The radius of third Bohr orbit of Li^{2+} ion is

(1) 1.587 pm (2) 158.7 Å
(3) 15.87 pm (4) 158.7 pm

59. Match List-I with List-II

	List-I (Molecules)		List-II (Shape)
(a)	BrF_5	(i)	Trigonal bipyramidal
(b)	PF_5	(ii)	Square pyramidal
(c)	SF_4	(iii)	Square planar
(d)	XeF_4	(iv)	See-saw

Choose the correct answer from the options given below.

- (1) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)
(2) (a)-(i), (b)-(iv), (c)-(iii), (d)-(ii)
(3) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)
(4) (a)-(ii), (b)-(i), (c)-(iii), (d)-(iv)
60. Which amongst the following is an **incorrect** statement?
- (1) H_2^+ has one electron
(2) Be_2 molecule does not exist
(3) Bond orders of O_2^+ , O_2 , and O_2^{2-} are 1.5, 2.5 and 2 respectively
(4) C_2 molecule has four electrons in its two degenerate π molecular orbitals
61. Extensive property among the following is
- (1) Internal energy (2) Temperature
(3) Density (4) Pressure
62. Consider the following statements
- (a) p-nitrophenol is more acidic than phenol.
(b) Oxidation of phenol with chromic acid produces a conjugated diketone.
(c) In Kolbe's reaction, phenol forms orthohydroxy benzaldehyde.

The correct statements are

(1) (a) and (b) only (2) (b) and (c) only
(3) (a) and (c) only (4) (a), (b) and (c)

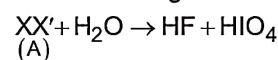
63. Mark the **incorrect** statement out of the following.

(1) Group I basic radicals precipitate as chlorides
(2) Group II basic radicals precipitate as sulphides
(3) Group V basic radicals precipitate as hydroxides
(4) Group IV basic radicals precipitate as sulphides

64. The species which will not undergo disproportionation reaction is

(1) S_8 (2) Cl_2
(3) P_4 (4) F_2

65. In the following reaction, reactant (A) is



(1) IF_3 (2) F_2
(3) IF_5 (4) IF_7

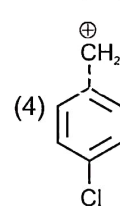
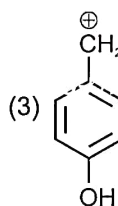
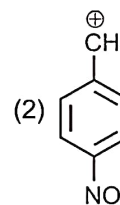
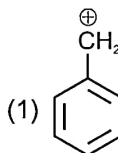
66. Glucose on reaction with bromine water gives

(1) Gluconic acid (2) Saccharic acid
(3) Glyceraldehyde (4) Sorbitol

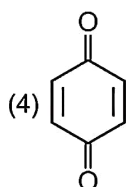
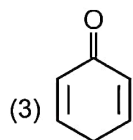
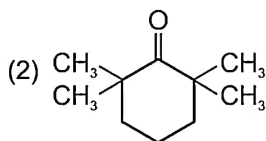
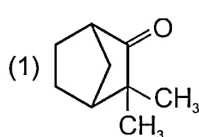
67. Which of the following gives positive Tollens' test but negative Fehling's test?

(1) HCHO (2) CH_3CHO
(3) Ph-CHO (4) CH_3COCH_3

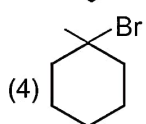
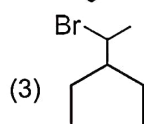
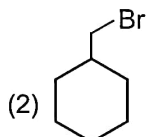
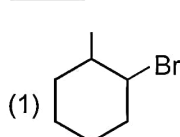
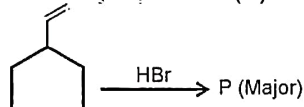
68. The least stable carbocation among the following is



69. Which among the following molecules can exhibit tautomerism?



70. The major product (P) obtained in the reaction is



71. Identify 'X' from the following reaction
 $X + K_4[Fe(CN)_6] \rightarrow \text{Prussian blue precipitate}$

- (1) $FeCl_3$ (2) $PbCl_2$
 (3) $ZnCl_2$ (4) $NiCl_2$

72. In Carius method of estimation of halogen, 0.28 g of an organic compound gave 0.2 g of $AgBr$. The percentage of bromine in the compound is

- (Atomic mass of $Ag = 108$ u, $Br = 80$ u)
 (1) 38.2% (2) 46.2%
 (3) 30.4% (4) 25.6%

73. 5% (w/v) solution of urea is isotonic with 2% (w/v) solution of a non-electrolyte substance. The molar mass of the substance is

- (1) 180 g/mol (2) 24 g/mol
 (3) 120 g/mol (4) 30 g/mol

74. Mole fraction of the solvent in a 1.00 molal aqueous solution is nearly

- (1) 0.0982 (2) 0.058
 (3) 0.982 (4) 0.0528

75. Match the reactions given in column-I with the statements given in column-II.

	Column-I		Column-II
a.	Ammonolysis	(i)	Preparation of amines with one carbon atom less than that present in the amide
b.	Gabriel phthalimide synthesis	(ii)	Detection test for primary amines
c.	Hoffmann bromamide degradation reaction	(iii)	Preparation of 1° aliphatic amines
d.	Carbylamine reaction	(iv)	Reaction of alkyl halides with NH_3

Choose the correct option.

- (1) a(i), b(ii), c(iii), d(iv)
 (2) a(iii), b(iv), c(i), d(ii)
 (3) a(iv), b(iii), c(i), d(ii)
 (4) a(iv), b(i), c(iii), d(ii)

76. The constituents of bronze are

- (1) Cu and Sn (2) Cu and Ag
 (3) Cu and Zn (4) Zn and Ag

77. The correct order of acidic character is

- (1) $H_2Te < H_2Se < H_2S < H_2O$
 (2) $H_2S < H_2O < H_2Se < H_2Te$
 (3) $H_2Se < H_2Te < H_2O < H_2S$
 (4) $H_2O < H_2S < H_2Se < H_2Te$

Space for Rough Work

78. Consider the following statements.

Statement I: The total entropy change (ΔS_{total}) for the system and surroundings of a spontaneous process is given by

$$\Delta S_{\text{total}} = \Delta S_{\text{system}} + \Delta S_{\text{surr}} > 0$$

Statement II: When a system is in equilibrium, the entropy is maximum, and the change in entropy, $\Delta S = 0$.

Choose the **correct** option.

- (1) Statement I is correct but statement II is incorrect
- (2) Statement I is incorrect but statement II is correct
- (3) Both statement I and statement II are incorrect
- (4) Both statement I and statement II are correct

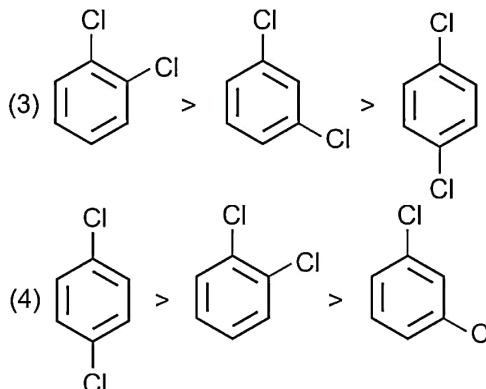
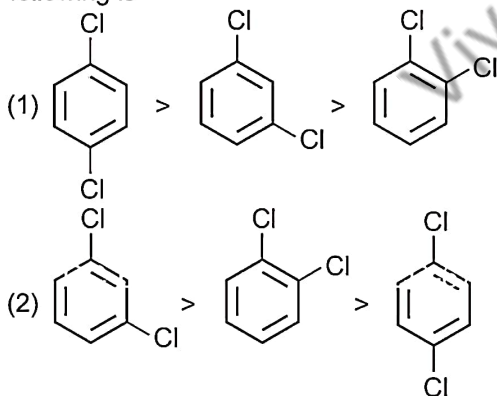
79. The number of moles of KMnO_4 needed to react completely with two moles of iodide ion in acidic medium is

- (1) $\frac{2}{5}$ moles
- (2) $\frac{1}{5}$ moles
- (3) $\frac{1}{3}$ moles
- (4) $\frac{2}{3}$ moles

80. The maximum number of ions produced from the complex $\text{K}_4[\text{Fe}(\text{CN})_6]$ in an aqueous solution is

- (1) 4
- (2) 5
- (3) 6
- (4) 7

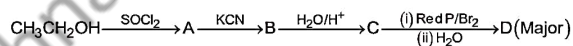
81. The **correct** order of melting point among the following is



82. Which of the following statements is incorrect about diborane?

- (1) The four terminal hydrogen atoms and the two Boron atoms lie in one plane
- (2) The bond length of terminal B-H bond is more than the bond length of bridge B-H bond
- (3) Both the Boron atoms are sp^3 hybridised
- (4) The four terminal B-H bonds are two centre two electrons bonds

83. Consider the following reaction,



product D is

- (1) $\text{CH}_3\text{CH}_2\text{C}(=\text{O})\text{Br}$
- (2) $\text{CH}_3\text{CH}(\text{Br})\text{COOH}$
- (3) $\text{CH}_3\text{CH}_2\text{CH}_2\text{Br}$
- (4) $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$

84. Consider the following statements.

Statement I: Primary aliphatic amines react with HNO_2 to give stable diazonium salts.

Statement II: Primary aromatic amines react with HNO_2 to form diazonium salts at low temperatures (273-278 K).

Choose the **correct** option.

- (1) Statement I is correct but statement II is incorrect
- (2) Statement I is incorrect but statement II is correct
- (3) Both statement I and statement II are incorrect
- (4) Both statement I and statement II are correct

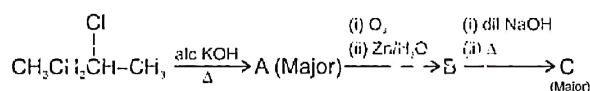
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85. The least acidic compound among the following is

- (1) CH_3COOH
- (2) $\text{C}_6\text{H}_5\text{CH}_2\text{COOH}$
- (3) HCOOH
- (4) $\text{C}_6\text{H}_5\text{COOH}$

SECTION-B

86. In the following reaction sequence



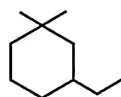
product C is

- (1) But-3-enal
- (2) But-2-enal
- (3) Butan-2-one
- (4) Butanoic acid

87. Which of the following is a free radical substitution reaction?

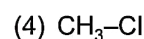
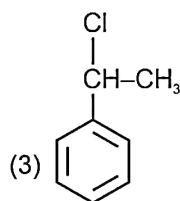
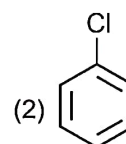
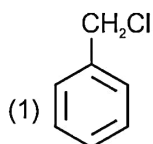
- (1) Propene with $\text{HBr}/(\text{C}_6\text{H}_5\text{CO})_2\text{O}_2$
- (2) Methane with $\text{Br}_2/h\nu$
- (3) Propyne with HgSO_4 and dil. H_2SO_4 at 333 K
- (4) Benzene with CH_3Cl in presence of anhydrous AlCl_3

88. Which of the following options represents correct IUPAC name for the given compound?



- (1) 1-Ethyl-3,3-dimethylcyclohexane
- (2) 3-Ethyl-1,1-diethylcyclohexane
- (3) 1, 1-Dimethyl-3-Ethylcyclohexane
- (4) 3-Ethyl-1, 1-dimethylcyclohexane

89. Which of the following is most reactive toward $\text{S}_{\text{N}}1$ reaction?



90. Which of the following statements is correct regarding a solution of two components A and B exhibiting negative deviation from ideal behaviour?

- (1) $\Delta_{\text{mix}}H \geq 0$ at constant T and P
- (2) Intermolecular attractive forces between A-A and B-B are stronger than those between A-B
- (3) $\Delta_{\text{mix}}V = 0$ at constant T and P
- (4) Intermolecular attractive forces between A-A and B-B are weaker than those between A-B

91. Consider the following statements.

Statement I: Lead storage battery consists of a lead anode and a grid of lead packed with lead dioxide as cathode.

Statement II: A 38% solution of hydrochloric acid is used as an electrolyte, in lead storage battery.

Choose the **correct** option.

- (1) Both statement I and statement II are correct
- (2) Statement I is correct but statement II is incorrect
- (3) Statement I is incorrect but statement II is correct
- (4) Both statement I and statement II are incorrect

Space for Rough Work

Test-12 (Code-A)

92. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A): Picric acid reacts with sodium bicarbonate to give effervescence of carbon dioxide.

Reason (R): Picric acid is a stronger acid than carbonic acid.

In the light of above statements, choose the correct option among the following.

- (1) (A) is true but (R) is false
 - (2) Both (A) and (R) are true and (R) is the correct explanation of (A)
 - (3) (A) is false but (R) is true
 - (4) Both (A) and (R) are true but (R) is not the correct explanation of (A)
93. For the reversible reaction,
- $$\text{X}_2(\text{g}) + 4\text{Y}_2(\text{g}) \rightleftharpoons 2\text{XY}_4(\text{g}) + \text{heat}$$
- The equilibrium shifts in forward direction
- (1) By increasing concentration of $\text{XY}_4(\text{g})$
 - (2) By decreasing the pressure
 - (3) By decreasing the concentration of $\text{X}_2(\text{g})$ and $\text{Y}_2(\text{g})$
 - (4) By increasing pressure and decreasing temperature
94. In the reaction,
- $$2\text{Mg}(\text{s}) + \text{O}_2(\text{g}) \rightarrow 2\text{MgO}(\text{s})$$
- When 1 g of magnesium and 0.32 g of oxygen are made to react to completion
- (1) 0.1 mole of MgO will be produced
 - (2) All the magnesium is consumed
 - (3) All the oxygen is consumed
 - (4) 0.03 moles of $\text{Mg}(\text{s})$ left
95. Consider the following statements.

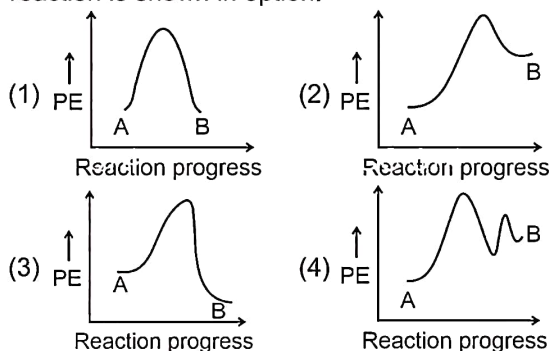
Statement I: The metal carbon bond in metal carbonyls possesses both σ and π character.

Statement II: The M-C π bond is formed by the donation of a pair of electrons from a filled d orbital of metal into the vacant antibonding π^* orbital of carbon monoxide.

Choose the correct option.

- (1) Both statement I and statement II are correct
- (2) Statement I is correct but statement II is incorrect
- (3) Statement I is incorrect but statement II is correct
- (4) Both statement I and statement II are incorrect

96. For a reaction $\text{A} \rightarrow \text{B}$, enthalpy of reaction is -4.6 kJ mol^{-1} and enthalpy of activation is 0.6 kJ mol^{-1} . The correct potential energy profile for the reaction is shown in option.



97. Pernicious anaemia is a deficiency disease of
- (1) Vitamin C
 - (2) Vitamin B_{12}
 - (3) Vitamin D
 - (4) Vitamin K
98. Group reagent for Al^{3+} cations is
- (1) H_2S in presence in NH_4OH
 - (2) H_2S gas in presence of dil. HCl
 - (3) NH_4OH in presence of NH_4Cl
 - (4) Dilute HCl
99. The molar solubility of CaF_2 ($K_{\text{sp}} = 5.3 \times 10^{-11}$) in 0.02 M solution of NaF will be
- (1) $1.32 \times 10^{-7} \text{ mol L}^{-1}$
 - (2) $5.3 \times 10^{-9} \text{ mol L}^{-1}$
 - (3) $1.32 \times 10^{-9} \text{ mol L}^{-1}$
 - (4) $5.3 \times 10^{-10} \text{ mol L}^{-1}$
100. Match List-I with List-II and choose the correct option.

	List-I (Compound)		List-II (Colour)
(a)	PbS	(i)	Dark yellow
(b)	$\text{Na}_4[\text{Fe}(\text{CN})_5\text{NOS}]$	(ii)	Black
(c)	AgI	(iii)	Red
(d)	$[\text{Ce}(\text{NO}_3)_4(\text{ROH})_3]$	(iv)	Violet

- (1) a(i), b(iii), c(ii), d(iv)
- (2) a(iii), b(i), c(ii), d(iv)
- (3) a(iv), b(ii), c(i), d(iii)
- (4) a(ii), b(iv), c(i), d(iii)

Space for Rough Work

101. Read the following statements.

- Vegetative reproduction takes place by fragmentation.
- Sexual spores are endogenously produced.
- Mycelium is unbranched and septate
- Sex organs are absent in them

Regarding mushroom the **correct** statements are

- a and c
- b, c and d
- a and d
- b and d only

102. Which of the following karyotypic conditions can lead to the disorder which causes gynaecomastia as one of the prominent features?

- 44 + XXY
- 44 + XY
- 44 + XO
- 44 + XYY

103. Which of the following criteria is **not** considered while identifying any areas as hotspots?

- They have very high degree of endemism
- They have very high level of species richness
- These regions are resistant to habitat loss
- They require certain measures to save it from extinction

104. Which process is performed in a template dependent manner?

- Tailing of hnRNA
- Capping of hnRNA
- RNA polymerisation by a polynucleotide phosphorylase
- Synthesis of hnRNA

105. False fruit

- Develops only from the ovary
- Develops without the process of fertilisation
- Are seedless and produced through application of auxin
- Develops from other floral parts along with ovary

106. Select the **incorrect** match

- Ethanol – Fungal product
- Cyclosporin A – Bacterial product
- Butyric acid – Produced by *Clostridium*
- Streptokinase – Used as clot buster

107. In which among the following cases, a child will exhibit heterozygous genotype for blood group A?

- Both the parents have same genotype, i.e., $I^A I^A$.
- Both the parents are of blood group O.
- One of the parent may have blood group AB while other have O.

- Only A
- B and C
- Only B
- A and C

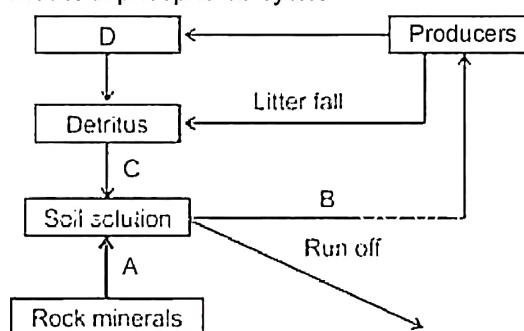
108. Read the following statements and choose the appropriate option.

Statement A : In C_3 plant, RuBisCO functions only as carboxylase.

Statement B : Photorespiration results in the release of CO_2 with the utilisation of ATP.

- Only statement A is correct
- Only statement B is correct
- Both the statement A and statement B are correct
- Both the statement A and statement B are incorrect

109. Select the **correct** option w.r.t. given below model of phosphorus cycle.



Space for Rough Work

- (1) Level 'D' is occupied by plants
 (2) 'C' is the process which takes place through decomposers
 (3) 'A' is the process by which uptake of phosphorus by plants occur
 (4) 'B' is the process through which gaseous form of phosphorus is absorbed by producers
110. Select the **incorrect** statement w.r.t. physiological effects of ethylene.
 (1) It increases the number of female flowers in cucumber
 (2) It accelerates abscission of plant parts
 (3) It inhibits sprouting of potato tubers
 (4) It breaks dormancy of seeds
111. Select the **incorrect** statement w.r.t. mesosomes.
 (1) They are formed by the extension of plasma membrane into the cell
 (2) These are present in the form of vesicles, tubules and lamellae
 (3) They contain photosynthetic pigments in some prokaryotes like cyanobacteria
 (4) They help in cell wall formation
112. The beginning of diplotene is recognised by which of the given events?
 (1) Bivalent formation
 (2) Crossing over
 (3) Terminalisation of chiasmata
 (4) Dissolution of synaptonemal complex
113. If a meiocyte has 18 chromosomes, how many bivalents will be formed during the process of meiosis I?
 (1) 9 (2) 18
 (3) 36 (4) 4
114. How many kingdoms w.r.t. Whittaker's classification system exhibit only autotrophic and only heterotrophic mode of nutrition respectively?
- (1) 1 and 3
 (2) 1 and 2
 (3) 2 and 1
 (4) 2 and 3
115. From which of the given regions of root-tip, root hairs develop?
 (1) Root cap
 (2) Region of meristematic activity
 (3) Region of elongation
 (4) Region of maturation
116. There is no secondary growth in monocot root, **because**
 (1) Of the presence of large pith
 (2) Vascular bundle is polyarch in them
 (3) Of the absence of cambium
 (4) Root never increases its girth
117. Select the **incorrect** statement w.r.t. photosystem II.
 (1) The reaction centre is P₆₈₀
 (2) PS II occurs only on the outer surface of stroma lamellae
 (3) Involved in non-cyclic photophosphorylation
 (4) Associated with splitting of water and release of O₂
118. Which of the given steps is catalysed by pacemaker enzyme w.r.t. EMP pathway?
 (1) Glucose to Glucose-6-phosphate
 (2) Fructose-6-phosphate to fructose-1, 6-bisphosphate
 (3) 2-phosphoglycerate to phosphoenolpyruvate
 (4) Glyceraldehyde-3-phosphate to 1, 3-bisphosphoglyceric acid
119. Select the **incorrect** match.
 (1) Tuber – Potato
 (2) Rhizome – Banana
 (3) Bulb – Onion
 (4) Phyllode – *Euphorbia*

Space for Rough Work

120. To fix every CO₂ molecule entering Calvin cycle, how many molecules of ATP and NADPH are required respectively?
- (1) 3 and 2 (2) 2 and 3
(3) 4 and 5 (4) 5 and 4
121. Consider the following statements.
Assertion (A): In the plant cell, the concentration of ions is significantly higher in the vacuole than in the cytoplasm.
Reason (R): The membrane of the vacuole facilitates the transport of ions, only along the concentration gradient into a vacuole.
In the light of above statements choose the correct option.
- (1) Both (A) and (R) are true and (R) is correct explanation of (A)
(2) Both (A) and (R) are true and (R) is not the correct explanation of (A)
(3) (A) is true and (R) is false
(4) Both (A) and (R) are false
122. Racemose inflorescence is characterised by
- (1) Limited growth of main axis
(2) Basipetal order of flowers borne on it
(3) Position of flower at the tip of main axis
(4) Presence of younger flowers towards the apex
123. Choose the statement which is **not** true for endodermis.
- (1) In roots, it comprises of barrel-shaped cells in which the cell walls have deposition of suberin
(2) It is present in both monocot stem and root
(3) It is innermost layer of cortex
(4) The cells of endodermis can store starch grains in dicot stem
124. All of the following are characteristic of flowers pollinated by insects or animals, **except**
- (1) They are large and colorful
(2) They provide rewards to pollinator
(3) They can be white and rich in nectar
(4) Non-sticky pollens are present in them
125. *i* gene of *lac* operon
- (1) Produces repressor protein which binds to RNA polymerase
(2) Synthesises β galactosidase
(3) Cannot express itself in absence of lactose
(4) Is a constitutive gene
126. DNA in the nucleus of a typical mammalian cell contains 6.6×10^9 bp. How many nucleosomes in the chromatin are possible?
- (1) 3.3×10^7
(2) 6.6×10^7
(3) 6.6×10^9
(4) 3.3×10^9
127. Match the following column w.r.t. sacred groves/lake with their locations and choose the correct option.
- | Column I | Column II |
|----------------------------|----------------|
| a. Khecheopalri lake | (i) Karnataka |
| b. Aravalli hills | (ii) Meghalaya |
| c. Western Ghats | (iii) Sikkim |
| d. Khasi and Jaintia hills | (iv) Rajasthan |
- (1) a(iv), b(i), c(ii), d(iii)
(2) a(iii), b(iv), c(i), d(ii)
(3) a(iii), b(ii), c(iv), d(i)
(4) a(iv), b(iii), c(i), d(ii)
128. The equation that represents geometric fashion of growth of a population is
- (1) $\frac{dN}{dt} = rN$
(2) $\frac{dN}{dt} = rN \left(\frac{K - N}{K} \right)$
(3) $\frac{dN}{dt} = rN \left(\frac{K}{K + N} \right)$
(4) $\frac{dN}{dt} = N^r$

129. Identify the **incorrect** statements about sickle cell anaemia and choose the option accordingly.
- This disorder is caused by a transition mutation.
 - This disorder is developed due to presence of two pairs of alleles on the same chromosome.
 - It is an autosomal recessive trait.
 - This defect is caused due to substitution of valine by glutamine in β -chain of haemoglobin molecule.
 - Individual with genotype $Hb^A Hb^S$ is normal but carrier.
- (1) ii, iii and iv
 - (2) i and iii
 - (3) iv and v
 - (4) i, ii and iv
130. Read the following statements
- Assertion (A):** At the end of meiosis II four haploid cells are formed from a diploid cell.
- Reason (R):** Meiosis involves pairing of homologous chromosomes and recombination between non-sister chromatids of homologous chromosomes.
- In the light of above statements, choose the **correct** option.
- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
 - (2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
 - (3) (A) is true and (R) is false
 - (4) Both (A) and (R) are false
131. Which of the given gametic fusion results nullisomic condition?
- (1) $n \times (n - 1)$
 - (2) $(n - 1) \times (n - 1)$
 - (3) $n \times (n + 1)$
 - (4) $(n + 1) \times (n + 1)$
132. Which of the given is **not** the characteristics of anthropogenic ecosystem?
- (1) Possess self regulatory mechanism
 - (2) Have little diversity
 - (3) Little cycling of nutrients
 - (4) Simple food chain
133. Select the **incorrect** match w.r.t. number of species of different vertebrates in Amazonian rain forest.
- (1) Mammals – 427
 - (2) Birds – 13000
 - (3) Reptiles – 378
 - (4) Fishes – 3000
134. **Assertion (A) :** When a person goes to high altitudes he experiences altitude sickness with symptoms like difficulty in breathing and heart palpitations.
- Reason (R) :** Due to low atmospheric pressure at high altitudes, the body does not get sufficient oxygen.
- In the light of above statements, choose the **correct** option.
- (1) Both (A) and (R) are false statements
 - (2) Only (R) is true statement
 - (3) Both (A) and (R) are true but (R) is not the correct explanation of (A)
 - (4) Both (A) and (R) are true and (R) is the correct explanation of (A)
135. In taxonomic hierarchy, the number of similar characters of the organisms belonging to the particular category decreases from
- (a) Kingdom to species
 - (b) Lowest to highest rank
 - (c) Highest to lowest rank
- The **correct** one(s) is/are:
- (1) Only (a)
 - (2) Both (a) and (b)
 - (3) Only (b)
 - (4) Both (a) and (c)

Space for Rough Work

136. For artificial hybridisation, there is no need of emasculation when
- (1) Female parent produces unisexual flowers
 - (2) Female parent produces bisexual flowers
 - (3) There is a synchrony between maturation of stamen and pistil in a flower
 - (4) Anthers and stigma lie close to each other in a flower
137. Lodicules are found in members of
- (1) Compositae
 - (2) Malvaceae
 - (3) Poaceae
 - (4) Fabaceae
138. Who performed series of experiments that revealed essential role of air in the growth of green plants?
- (1) Cornelius van Niel
 - (2) Jan Ingenhousz
 - (3) Julius von Sachs
 - (4) Joseph Priestley
139. Choose the **incorrect** statement among them.
- (1) Stomata regulate gaseous exchange in plants.
 - (2) Cuticle is present on both stem and roots of all plants
 - (3) Mesophyll cell possess chloroplast and carry out photosynthesis
 - (4) Cork cambium is involved in secondary growth
140. Select the **incorrect** statement w.r.t. humus.
- (1) It is formed by the process of decomposition of detritus
 - (2) It is light coloured amorphous substance
 - (3) It is slightly acidic, colloidal and acts as reservoir of nutrients
 - (4) It is highly resistant to microbial action
141. Mark the **incorrect** statement regarding the features of human genome.
- (1) The human genome contains 3164.7 million bp
 - (2) The DNA sequences that are repeated many times have direct coding functions in genome.
 - (3) Less than 2% of the genome codes for proteins
 - (4) The average gene consists of 3000 bases, but the size vary greatly
142. The physical process w.r.t treatment of waste water involves
- (A) Removal of floating debris
 - (B) Formation of flocs in aeration tank
 - (C) Digestion of bacteria and fungi present in sludge in anaerobic sludge digester
 - (D) Removal of soil and small pebbles
- The **correct** ones are
- (1) A and B
 - (2) B and C
 - (3) C and D
 - (4) A and D
143. Mark the **incorrect** statement for the members of algae which have mannitol and laminarin as stored food?
- (1) They are brown algae
 - (2) They contain chlorophyll a, c and fucoxanthin as major pigments
 - (3) They have cellulose, pectin and polysulphate esters in their cell wall
 - (4) Their zoospores are characterised by presence of two, unequal, lateral flagella
144. Which of the given statements is **not** true for stenothermal organisms?
- (1) They live in areas where the temperature is uniform throughout the year
 - (2) They cannot tolerate large temperature variations
 - (3) They include most of the mammals and birds
 - (4) They are restricted to narrow range of temperatures.
145. **Assertion (A)** : *Funaria* exhibit haplo-diplontic life cycle pattern.
- Reason (R)** : In bryophytes, the dominant phase is represented by sporophyte which is independent, photosynthetic vascular plant body. In the light of above statements, choose the **correct** option.
- (1) Both (A) and (R) are true and (R) is correct explanation of (A)
 - (2) Both (A) and (R) are true and (R) is not correct explanation of (A)
 - (3) Both (A) and (R) are false statements
 - (4) Only (A) is true statement

Space for Rough Work

146. The parasitic fungi causing rust disease on mustard is

- (1) *Albugo* (2) *Phytophthora*
(3) *Puccinia* (4) *Claviceps*

147. Stem tendrils occur in

- (a) Watermelon
(b) Cucumber
(c) Pumpkin

The **correct** one(s) is/are

- (1) (a) only (2) (a) and (b) only
(3) (b) and (c) only (4) All (a), (b) and (c)

148. In Mendelian dihybrid cross for seed shape and seed colour, the ratio amongst genotypes RrYy, rrYy, Rrry and rrry in F₂ generation is

- (1) 4 : 2 : 2 : 1 (2) 2 : 4 : 2 : 1
(3) 2 : 2 : 4 : 1 (4) 1 : 2 : 2 : 4

149. Which among the following are inclusion bodies in blue green algae?

- (1) Centrioles
(2) Microtubules
(3) Contractile vacuoles
(4) Gas vacuoles

150. If a meiocyte has 24 chromosomes with 10 pg DNA (at G₁ phase) then what will be the total number of chromatids and amount of DNA in a meiosis-I product?

	Chromatids	DNA(in pg)
(1)	48	10
(2)	24	5
(3)	24	10
(4)	24	20

SECTION - A

151. The epithelium which has a limited role in secretion and absorption is present in

- (1) Lining of PCT
(2) Lining of stomach
(3) Moist surface of buccal cavity
(4) Inner lining of bronchioles

152. The beginning of atrial diastole coincides with the

- (1) End of ventricular systole
(2) Beginning of ventricular diastole
(3) Beginning of ventricular systole
(4) Beginning of atrial systole

153. A girl 'X' was walking calmly down the road when all of sudden a snake appeared in front of her. The following physiological changes would have take place in her body, **except**

- (1) Excessive sweating
(2) Dilation of pupils
(3) Increase in heart rate
(4) Decrease in cardiac output

154. Select the **incorrect** statement regarding regulation of respiration in humans.

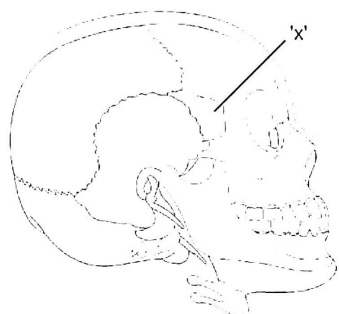
- (1) Both chemosensitive area and the respiratory rhythm centre are present in medulla oblongata.
(2) Change in CO₂ and H⁺ concentration can be recognised by receptors present in the aortic arch.
(3) The role of oxygen in regulation of the respiratory rhythm is most significant.
(4) Pneumotaxic centre is present in the pons region of the brain.

Space for Rough Work

155. Which of the following is **incorrect** for cloning vector pBR322?
- (1) It has ampicillin resistance gene.
 - (2) It has a special sequence called 'ori'.
 - (3) It has specific recognition sites for the different restriction enzymes.
 - (4) Selection of recombinants is always a single step process while using it.
156. Consider the given statements and select the **correct** option.
- Statement (A):** Heart of both humans as well as amphibians is neurogenic in nature.
- Statement (B):** Neural signals through the sympathetic nerve fibres increase the strength of ventricular contraction in humans.
- (1) Both statements (A) and (B) are correct
 - (2) Both statements (A) and (B) are incorrect
 - (3) Only statement (A) is incorrect
 - (4) Only statement (B) is incorrect
157. Select the option that represents **correct** set of bones included under appendicular skeleton of humans.
- (1) Coxal bone, Sternum, Coccyx
 - (2) Patella, Ischium, Scapula
 - (3) Radius, Ulna, Atlas
 - (4) Axis, Malleus, Stapes
158. *Bt* toxin gene has been introduced in plants in order to provide resistance against
- (a) Tobacco budworm and armyworm
 - (b) Beetles and corn borers
 - (c) Flies and mosquitoes
- Choose the **correct** option.
- (1) Only (a) and (b)
 - (2) Only (b) and (c)
 - (3) Only (a) and (c)
 - (4) (a), (b) and (c)
159. The anatomical unit of muscle is
- (1) Muscle fibre
 - (2) Sarcomere
 - (3) Myosin filament
 - (4) Actin filament
160. Choose the **incorrect** statement.
- (1) Sucrose is a disaccharide made up of glucose and fructose.
 - (2) Lecithin is a phosphorylated glyceride found in the cell membranes.
 - (3) Only peptide bonds are involved in stabilizing tertiary structure of the proteins.
 - (4) Lipids are not polymeric compounds.
161. Which of the following is characterised by three-horned face and quadrupedal locomotion?
- (1) *Tyrannosaurus*
 - (2) *Triceratops*
 - (3) *Stegosaurus*
 - (4) *Brachiosaurus*
162. Select the **correct** set of animals that belong to the same phylum.
- (1) *Sycon*, *Spongilla*, *Scoliodon*
 - (2) *Physalia*, *Pleurobrachia*, *Pennatula*
 - (3) *Columba*, *Canis*, *Ctenoplane*
 - (4) *Anopheles*, *Aedes*, *Apis*
163. How many of the organisms given in the box below are uricotelic in nature?
- Reptiles, Birds, Cartilaginous fishes, Aquatic insects, *Hydra*, Land snails
- Select the **correct** option
- (1) Three
 - (2) Four
 - (3) Five
 - (4) Six

Space for Rough Work

164. Which of the following does not hold true for the bone marked as 'X' in the figure given below?



- (1) It is an unpaired bone.
 (2) It is a cranial bone.
 (3) It is a facial bone.
 (4) It is a part of axial skeleton.
165. Which one of the following is **not** found in renal pyramid?
- (1) Collecting ducts (2) Loop of Henle
 (3) Vasa recta (4) Convoluted tubules
166. Match column I with column II w.r.t. humans and choose the **correct** option.

Column I	Column II
a. Hinge joint	(i) Between humerus and pectoral girdle
b. Pivot joint	(ii) Between carpal and metacarpal of thumb
c. Ball and socket joint	(iii) Between femur and tibia
d. Saddle joint	(iv) Between atlas and axis

- (1) a(i), b(ii), c(iii), d(iv)
 (2) a(iii), b(iv), c(i), d(ii)
 (3) a(iv), b(iii), c(ii), d(i)
 (4) a(i), b(iii), c(iv), d(ii)

167. Which of the following is **correct** w.r.t. repolarisation of axonal membrane?

- (1) Opening of voltage gated Na^+ channels
 (2) Closure of voltage gated Na^+ channels and opening of voltage gated K^+ channels
 (3) Closure of voltage gated K^+ channels
 (4) Closure of both voltage gated Na^+ and K^+ channels

168. All of the following ARTs involve *in vitro* fertilisation, **except**

- (1) ZIFT
 (2) ICSI
 (3) IUT
 (4) GIFT

169. Which of the following is **true** for organisms belonging to the class 'Aves'?

- (1) They have no glands
 (2) They are poikilothermous
 (3) Their forelimbs have scales and are modified for walking and clasping the tree branches
 (4) They have hollow long bones with air cavities present in them

170. The total number of disulphide bonds present in a molecule of monomeric antibody is

- (1) Ten
 (2) Sixteen
 (3) Fourteen
 (4) Twelve

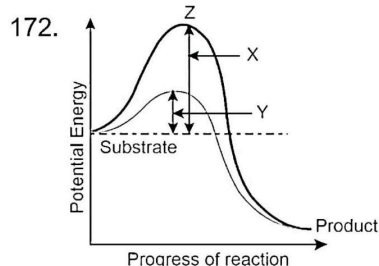
171. How many of the compounds given in the box exist(s) as quaternary structure of proteins and formed by two subunits of α -type and two subunits of β -type?

Myoglobin, Haemoglobin	Collagen,	Insulin,	Myosin,
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Select the **correct** option.

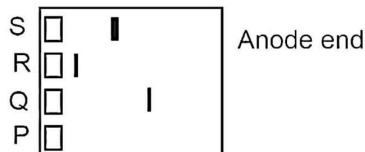
- (1) Three (2) One
 (3) Four (4) Five

Space for Rough Work



With respect to above graph, select the **correct** option.

- (1) 'Y' represents activation energy without an enzyme
 - (2) 'Z' represents a stable and lowest energy state
 - (3) 'X' represents activation energy in the presence of an enzyme
 - (4) This graph represents an exothermic reaction
173. Choose the option which represents the **correct** sequence of passage of sperms in male frog.
- (1) Testes → Bidder's canal → Vasa efferentia → Urinogenital duct → Cloaca
 - (2) Testes → Vasa efferentia → Bidder's canal → Urinogenital duct → Cloaca
 - (3) Testes → Bidder's canal → Vasa efferentia → Rectum → Urinogenital duct
 - (4) Testes → Vasa efferentia → Urinogenital duct → Bidder's canal → Cloaca
174. Although frogs have different modes of respiration but the structure commonly used by them in water, land and in unfavourable situations is
- (1) Lungs
 - (2) Skin
 - (3) Buccal cavity
 - (4) Lungs and buccal cavity
175. In an experimental set up, three samples were taken. Sample 'A' has DNA fragments, each with size of 8 kb. Sample 'B' has DNA fragments each with size of 2 kb and sample 'C' has DNA fragments, each with size of 4 kb. These samples were subjected to gel electrophoresis and the result obtained is given below.



Which of the above lanes (P, Q, R, S) represents DNA fragments of sample 'A'?

- (1) Q
 - (2) S
 - (3) P
 - (4) R
176. A biological response modifier activates the immune system and helps in destroying the tumor. It belongs to the _____ barrier of innate immunity.
- Select the **correct** option to fill in the blank.
- (1) Physical
 - (2) Cellular
 - (3) Physiological
 - (4) Cytokine
177. **Assertion (A):** In recombinant DNA technology, both the vector DNA and DNA of interest are cut by the same restriction endonuclease.
- Reason (R):** The same kinds of sticky ends are required to facilitate the action of DNA ligase.
- In the light of above statements, select the **correct** option.
- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
 - (2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
 - (3) (A) is true, (R) is false
 - (4) Both (A) and (R) are false
178. For transformation of a bacterial cell, it has to be made competent because
- (1) Competent cells have modified DNA in their genome which help them to transform easily.
 - (2) Competent cells are able to take alien piece of DNA efficiently.
 - (3) Competent cells are able to take up the alien piece of DNA, modify and kill it.
 - (4) Competency keeps the cell wall rigid.

Space for Rough Work

179. Vaccine which contains weakened pathogens or antigenic preparations provides
- (1) Natural active immunity
 - (2) Natural passive immunity
 - (3) Artificial active immunity
 - (4) Artificial passive immunity
180. All of the following statements are true w.r.t. tubectomy, **except**
- (1) It only blocks gamete transport not its production.
 - (2) Follicles within ovary develop normally.
 - (3) Reversibility is very poor.
 - (4) Females who undergo tubectomy do not menstruate
181. Select the **correct** match.
- (1) 50% of human sperms – Contain 22 + X chromosomes
 - (2) Primary spermatocyte – Haploid cell
 - (3) Secondary oocyte – Diploid cell
 - (4) Spermatids – Contain flagella and are motile
182. If a human female has a 28-day menstrual cycle, then the highest level of estrogen in blood will be seen from
- (1) Day 5 to Day 9
 - (2) Day 13 to Day 14
 - (3) Day 17 to Day 19
 - (4) Day 3 to Day 7
183. Choose the **incorrect** statement w.r.t. human evolution.
- (1) The skull of baby chimpanzee is more like adult chimpanzee skull than adult human skull.
 - (2) The story of evolution of modern man appears to parallel evolution of human brain and language.
 - (3) Agriculture came around 10,000 years back and human settlements started.
 - (4) *Ramapithecus* was more man-like while *Dryopithecus* was more ape-like.
184. Which of the following structures allows passage of small amounts of urea into medullary interstitium to keep up the osmolarity?
- (1) Descending limb of Henle's loop
 - (2) Collecting duct
 - (3) Ascending limb of Henle's loop
 - (4) Distal convoluted tubule
185. For maintenance of osmolarity of body fluids, descending limb of loop of Henle _____ the filtrate whereas ascending limb of loop of Henle _____ the filtrate majorly.
- Select the **correct** option to fill in the blanks respectively.
- (1) Dilutes ; Concentrates
 - (2) Concentrates ; Dilutes
 - (3) Reabsorbs ; Secretes
 - (4) Secretes ; Reabsorbs
- SECTION - B**
186. AV node delays impulses transmitted from SAN by 0.1 second so that
- (1) Ventricles can empty completely
 - (2) The atria can empty completely
 - (3) The atria get filled completely
 - (4) Half the blood present in the atria can enter into ventricles
187. Cerebrum wraps around 'X' present in human brain which is a major coordinating centre for sensory and motor signalling.
- Identify 'X' and select the **correct** option.
- (1) Cerebellum
 - (2) Thalamus
 - (3) Hypothalamus
 - (4) Pons

Space for Rough Work

188. Select the statement which is **true** for normal inspiration in humans.

- (1) A dome shaped structure present on the lower side of thoracic chamber contracts and increases the volume of thoracic cavity in the dorso-ventral axis.
- (2) External inter-costal muscles lift up the ribs and the sternum causing an increase in the volume of thoracic chamber in the antero-posterior axis.
- (3) Volume of thoracic cavity increases which decreases the intra-pulmonary pressure less than the atmospheric pressure.
- (4) Air enters the lungs *via* passive process.

189. Which of the following factors favour the formation of oxyhaemoglobin?

- (1) High pO_2 , High pCO_2 , Low pH, Higher temperature
- (2) High pO_2 , Low pCO_2 , High pH, Lower temperature
- (3) Low pO_2 , High pCO_2 , High pH, Higher temperature
- (4) High pO_2 , Low pCO_2 , Low pH, Higher temperature

190. Flow of blood through the below mentioned pathway in humans is included under

Left ventricle $\xrightarrow{\text{Aorta}}$ Body tissues $\xrightarrow[\text{Cava}]{\text{Vena}}$ Right atrium

Choose the **correct** option.

- (1) Coronary circulation
- (2) Pulmonary circulation
- (3) Lymphatic circulation
- (4) Systemic circulation

191. Identify the organism having following features.

- (i) Endoskeleton made of calcareous ossicles
- (ii) Adults are radially symmetrical
- (iii) Presence of complex digestive system
- (iv) Presence of water vascular system

Select the **correct** option.

- (1) *Sycon*
- (2) *Antedon*
- (3) *Pila*
- (4) *Pleurobrachia*

192. Read the following statements.

Statement A: Zona pellucida is a non-cellular layer secreted by the secondary oocyte.

Statement B: Zona pellucida usually remains intact until the embryo reaches the uterus.

Choose the **correct** option.

- (1) Only statement A is correct
- (2) Only statement B is correct
- (3) Both statements A and B are incorrect
- (4) Both statements A and B are correct

193. The most primitive craniates from the options given below are

- (1) Cephalochordates
- (2) Urochordates
- (3) Cyclostomes
- (4) Gnathostomes

194. Choose the **incorrect** match regarding the parts of reproductive system of a male cockroach and their functions.

- | | | |
|---------------------------|---|------------------------------------|
| (1) Seminal vesicles | – | Secrete outer layer of spermatheca |
| (2) Mushroom shaped gland | – | Accessory reproductive gland |
| (3) Testes | – | Formation of sperms |
| (4) Vas deferens | – | Transportation of sperms |

195. Human insulin has been produced by bacterial cells in recombinant DNA technology.

Which special feature of genetic code is particularly responsible for the successful production of human insulin in bacteria?

- (1) Unambiguous nature of genetic code
- (2) Genetic code is degenerate
- (3) Genetic code is nearly universal
- (4) Genetic code is specific

Space for Rough Work

196. Amniocentesis is used to diagnose
 (1) Osteoporosis (2) Sickle cell anaemia
 (3) Myasthenia gravis (4) Arthritis

197. Select the **correct** match.

	Hormone	Source	Function
(1)	PTH	Parathyroid gland	Decreases the blood calcium levels
(2)	Thymosin	Thyroid gland	Differentiation of T-lymphocytes
(3)	Melatonin	Pineal gland	Regulate diurnal rhythm of body
(4)	Glucagon	α - cells of Islets of Langerhans	Decreases blood glucose levels

198. Complete the analogy by selecting the **correct** option.

Flippers of Penguins and Dolphins: Convergent evolution : : Thorns of *Bougainvillea* and tendrils of *Cucurbita* : _____

- (1) Parallel evolution
 (2) Divergent evolution
 (3) Cyclical evolution
 (4) Chemical evolution

199. In RNA interference technique, the advantage of introducing nematode specific genes into the tobacco plant is that

- (1) *Agrobacterium* cannot survive in this transgenic tobacco plant
 (2) Nematode specific genes produce both sense and anti-sense RNA in the host plant which terminates RNAi
 (3) Specific DNA of the nematode is silenced by the formation of ssRNA.
 (4) *Meloidogyne incognita* cannot survive in this transgenic tobacco plant

200. Choose the **incorrect** statement w.r.t. steroid hormones.

- (1) They have intracellular receptors.
 (2) Upon binding with their receptor, they regulate gene expression.
 (3) They bring about physiological response through cAMP and IP_3 .
 (4) Testosterone, cortisol, estrogen and progesterone are some of the examples

Space for Rough Work